1.(a)	(i) $a_2 = 1$	B1
	(ii) $a_{107} = 3$	B1 **(2)**
(b)	$\sum_{n=1}^{200}\left(2a_n - 1\right) = 5+1+5+1+.....5+1 = 100\times\left(5+1\right)$	M1
	$= 600$	A1 **(2)**
		(4 marks)

2.(a)	Attempts $(x \pm 2)^2 + (y \pm 5)^2 ..... = 0$	M1
	(i) Centre $(-2, 5)$	A1
	(ii) Radius $\sqrt{50}$ or $5\sqrt{2}$	B1
		(3)
(b)	Gradient of radius = $\frac{(5)-4}{(-2)-5} = -\frac{1}{7}$ which needs to be in simplest form	B1ft
	Uses $m_2 = -\frac{1}{m_1}$ to find gradient of tangent	M1
	Equation of tangent $y - 4 = "7"(x-5) \Rightarrow y = 7x - 31$	M1 A1
		(4)
		(7 marks)

3. (i)	$(x-4)^2 \geqslant 2x-9 \Rightarrow x^2-10x+25 \ldots 0$ $\Rightarrow (x-5)^2 \ldots 0$ Explains that "square numbers are greater than or equal to zero" hence (as $x \in \mathbb{R}$), $\Rightarrow (x-4)^2 \geqslant 2x-9$ *	M1 A1 A1* **(3)**
(ii)	Shows that it is not true for a value of *n* Eg. When $n=3$, $2^n+1=8+1=9$ × Not prime	B1 **(1)** **(4 marks)**

4.(a)	$\left(2-\frac{1}{4}x\right)^6 = 2^6, +{}^6C_1 2^5\left(-\frac{1}{4}x\right)^1 + {}^6C_2 2^4\left(-\frac{1}{4}x\right)^2 + {}^6C_3 2^3\left(-\frac{1}{4}x\right)^3 + \ldots$ $= 64 - 48x + 15x^2 - 2.5x^3$	B1, M1 A1 A1 **(4)**
(b)	$\left(2-\frac{1}{4}x\right)^6 + \left(2+\frac{1}{4}x\right)^6 = \left(64-48x+15x^2-2.5x^3\right) + \left(64+48x+15x^2+2.5x^3\right)$ $\approx 128 + 30x^2$	M1 B1ft A1 **(3)** **(7 marks)**

5.(a)	$\frac{dP}{dx} = 12 - \frac{3}{2}x^{\frac{1}{2}}$	M1A1
	Sets $\frac{dP}{dx} = 0 \rightarrow 12 - \frac{3}{2}x^{\frac{1}{2}} = 0 \rightarrow x^n = \ldots$	dM1
	$x = 64$	A1
	When $x = 64 \Rightarrow P = 12 \times 64 - 64^{\frac{3}{2}} - 120 = \ldots$	M1
	Profit = (£) 136 000	A1
		(6)
(b)	$\left(\frac{d^2P}{dx^2}\right) = -\frac{3}{4}x^{-\frac{1}{2}}$ and substitutes in their $x = 64$ to find its value or state its sign	M1
	At $x = 64$ $\frac{d^2P}{dx^2} = -0.09375 < 0 \Rightarrow$ maximum	A1
		(2)
		(8 marks)

6.(a)	Sets $f(3)=0 \rightarrow$ equation in k Eg. $27k-135-96-12=0$ $\Rightarrow 27k=243 \Rightarrow k=9$ * (= 0 must be seen)	M1 A1* **(2)**
(b)	$9x^3-15x^2-32x-12=(x-3)(9x^2+12x+4)$ $=(x-3)(3x+2)^2$	M1 A1 dM1 A1 **(4)**
(c)	Attempts $\cos\theta=-\frac{2}{3}$ $\theta=131.8°, 228.2°$ (awrt)	M1 A1 **(2)** **(8 marks)**

7.(a)	Attempts to use $31\,500 = 16\,200 + 9d$ to find 'd'	M1
	For 16 200 + their $d = (1700)$ where d has been found by an allowable method	M1
	Year 2 salary is (£)17 900	A1
		(3)
(b)	Attempts to use $31\,500 = 16\,200r^{9}$ to find 'r'	M1
	For 16 2000 × their $r = (1.077)$ where r has been found by an allowable method	M1
	Year 2 salary in the range $17440 \leqslant \mathbb{S} \leqslant 17450$	A1
		(3)
(c)	Attempts $\frac{10}{2}\{16200 + 31500\}$ or $\frac{16200\left(1.077^{10} - 1\right)}{1.077 - 1}$	M1
	Finds $\pm\left(\frac{10}{2}\{16200 + 31500\} - \frac{16200\left('1.077'^{10} - 1\right)}{'1.077' - 1}\right)$	dM1
	Difference = £7480 cao	A1
		(3)
		(9 marks)

8.(i)

$8^{2x+1}=6 \Rightarrow 2x+1=\log_8 6$ M1

$\Rightarrow 2x+1=\dfrac{\log_2 6}{\log_2 8}$ A1

$\Rightarrow 2x+1=\dfrac{\log_2 2+\log_2 3}{3}$ M1

$\Rightarrow x=\dfrac{\log_2 3}{6}-\dfrac{1}{3}$ A1

OR

$2^{6x+3}=6$

$\Rightarrow (6x+3)\log_2 2=\log_2 6$ M1 A1

$\Rightarrow (6x+3)=\log_2 2+\log_2 3$ M1

$\Rightarrow x=\dfrac{\log_2 3}{6}-\dfrac{1}{3}$ A1

(4)

(ii)

$\log_5(7-2y)=2\log_5(y+1)-1$

$\log_5(7-2y)=\log_5(y+1)^2-1$

$\log_5(7-2y)=\log_5(y+1)^2-\log_5 5$

$(7-2y)=\dfrac{(y+1)^2}{5}$

$y^2+12y-34=0 \Rightarrow y=$

OR

$2\log_5(y+1)-\log_5(7-2y)=1$

$\log_5(y+1)^2-\log_5(7-2y)=1$ M1

$\log_5\dfrac{(y+1)^2}{(7-2y)}=1$ dM1

$\dfrac{(y+1)^2}{(7-2y)}=5$ A1

$y^2+12y-34=0 \Rightarrow y=$ ddM1

$y=-6+\sqrt{70}$ oe only A1

(5)

(9 marks)

9 (a)	Uses $\tan\theta = \dfrac{\sin\theta}{\cos\theta} \rightarrow$ $\cos\theta - 1 = 4\sin\theta \dfrac{\sin\theta}{\cos\theta}$	M1
	$\cos^2\theta - \cos\theta = 4\sin^2\theta$ oe	A1
	Uses $\sin^2\theta = 1 - \cos^2\theta \rightarrow$ $\cos^2\theta - \cos\theta = 4\left(1 - \cos^2\theta\right)$	M1
	$5\cos^2\theta - \cos\theta - 4 = 0$ *	A1 *
		(4)
(b)	$(5\cos 2x + 4)(\cos 2x - 1) = 0$	M1
	Critical values of $-\dfrac{4}{5}, 1$	A1
	Correct method to find x from their $\cos 2x = -\dfrac{4}{5}$	dM1
	$x = 0, 1.25$	A1
		(4)
		(8 marks)

10 (a)	$\left(f'(x)\right) = -\frac{72}{x^3} + 2$	M1 A1
	Attempts to solve $f'(x) = 0 \Rightarrow x = ...$ via $x^{\pm n} = k, \quad k > 0$	dM1
	$x > \sqrt[3]{36}$ oe	A1
		(4)
(b)	$\int \frac{36}{x^2} + 2x - 13 \, dx = -\frac{36}{x} + x^2 - 13x \, (+c)$	M1 A1
	Uses limits 9 and 2 $= \left(-\frac{36}{9} + 9^2 - 13 \times 9\right) - \left(-\frac{36}{2} + 2^2 - 13 \times 2\right) = 0$ *	dM1 A1*
		(4)
(c)(i)	8	B1
(ii)	$\int_2^6 \left(\frac{36}{x^2} + 2x + k\right) dx = 0 \Rightarrow \left[-\frac{36}{x} + x^2 + kx\right]_2^6 = 0 \Rightarrow (30 + 6k) - (-14 + 2k) = 0$	
	$44 + 4k = 0 \Rightarrow k = -11$	M1 A1
		(3)
		(11 marks)